



upShaala

Applied Data Science With Generative AI

**Learn the skill of Generative AI in Certified Data Science courses.
Utilise cutting-edge AI approaches to unlock your creative potential.**



Opportunities to get paid while completing the course by mentoring and submitting projects in real-time to tier-1 businesses.



Join Upshaala International Internship Programme (IIP) in Singapore/UAE (online)



Work as a Data Science & AI Consultant at IIP in Singapore/UAE (online programme).



Obtain certification from Microsoft, IBM, or Google (internationally recognised).



Guaranteed interviews and resume shortlisting for tier-1 to tier-5 companies.

www.upshaala.com

The Industry Gap

The real industry demands expertise and real-time knowledge supplemented by the appropriate skillsets. The pillars of Upshaala are constructed on the concept of industry experience and practical learning. We are here to fill the gap created between fresh graduates and industrial job creation. Our courses offer live brand projects, industry expert mentoring and job placement training to catapult our student community into top-tier jobs with a portfolio that sets them apart from the rest.



Your future starts here!

It's easy to assume that the future is somewhere 'out there',
but it's actually being created right here & now!

Scope of UpShaala

The gambit of Data Science is taking over every other pillar of the business world. Every start-up and job description entails you harnessing the expertise of the Data Science world. Upshaala ensures its students get the top course material and training programs after thorough research about the opportunities that we can provide to our student community.

With our courses, we push our students to fill the wide gap in the business community by becoming experts in the field of Data Science.

- **80% industry experts say Data Science is essential for innovation.**
- **61% industry experts say lack of training is the biggest barrier to improving data analysis.**
- **60% to 70% of all Business Data goes unused. A professional training will allow analysts to better utilize this data.**



Job Opportunities

Upshaala is not only here to enrich our students with the expertise of Data Science but also to grasp tremendous job opportunities with the best salary turnovers. With the increasing scope in this field and an estimate of 14% job growth rate by the year 2030, we must educate our student community about the best opportunities. Our professional real-time training courses complemented with 100% job assurance aid success at a faster pace.

Here is what a Upshaala professional certification brings to you:

Designation	Roles & Responsibilities
Machine Learning Engineer	A machine learning engineer is responsible for developing and deploying machine learning models to solve complex problems and enhance system functionalities.
NLP Engineer	A Natural Language Processing (NLP) Engineer focuses on developing and implementing applications that enable computers to understand, interpret, and generate human language.
Deep Learning Engineer	A Deep Learning Engineer specializes in designing, developing, and deploying deep neural networks to solve complex problems across various domains.
Generative AI Developer	A Generative AI Developer specializes in creating algorithms and systems that generate new content, such as images, text, or audio, using generative models.
Data Analyst	A Data Analyst is responsible for collecting, processing, and analyzing data to provide valuable insights and support informed decision-making within an organization.
Chat GPT Expert	A Chat CPT (Chat Completion) Expert specializes in the development and optimization of chat-based conversational models, ensuring natural and contextually relevant interactions.

Why UpShaala?

Online learning can often lack enjoyment, interactivity, and depth, but upshaala is distinct in this regard.

- **Live Training Sessions from Industry Experts**

Expand your horizons beyond us by tapping into the wealth of knowledge from our peers and mentors. Each week, you'll have the exclusive opportunity to soak up valuable insights from experts who have left their mark in the industry.

- **Real world Project Experience**

Real world projects offer invaluable benefits for students, fostering a practical learning environment that complements theoretical knowledge with hand-on experience.

By actively engaging in such projects, students gain a deeper understanding of industry dynamics and challenges, allowing them to apply classroom concepts to real world scenarios.

- **Course Completion & Project Experience Certificate**

Upon successful fulfillment of all course requirements and objectives, each student will be awarded a comprehensive Course Completion Certificate. This validates your commitment to learning and growth. You'll receive a Project Experience Certificate, acknowledging your hands on contribution and practical skills in project management.

- **360° Placement Trainings**

Our 360° Placement Trainings offer a comprehensive learning experience, blending theory with practical workshops, mock interviews, and personality development sessions. Delivered by industry experts, this program equips students with the necessary skills and soft abilities to excel in their chosen careers.

- **Internship opportunities & Placement Assistance**

We offer impactful internships to provide practical industry exposure, complementing academic learning.

Additionally, our dedicated team provides comprehensive placement assistance, including resume workshops, interview preparation, and networking opportunities, to help students secure fulfilling roles in their chosen fields. Rest assured, we are here for you. We host live sessions to promptly address any questions you may have in real time.

Why should you learn Data Science?

Learning data science is like having a superpower in today's world. It's about understanding the secrets hidden in mountains of data all around us. When you learn data science, you gain the ability to make sense of this information, finding patterns and insights that can help businesses make smarter decisions. Plus, it's a ticket to a world of exciting job opportunities because almost every industry now needs people who can work with data to help them grow and succeed. So, learning data science isn't just about understanding numbers; it's about unlocking a whole new realm of possibilities for your career and for making a real impact in the world.

Data Analysts are in demand in these industries:



Government



Financial Services



Education



Manufacturing



Healthcare



Retail

Upon course completion, learners will be eligible for the following job roles:

Data Analyst

NLP Engineer

Deep Learning Engineer

Data Engineer

Generative AI Developer

Data Scientist

Data Architect

Chat GPT Expert

AI Research Scientist

Machine Learning Engineer

This program caters to individuals who are:

Data Analyst

Quality Analyst

MIS Analyst

Financial Analyst

Customer Support Representative

Engineer

Consultant

Student

Business Leads

Developer

Our Advisors

Raj Kapadia

ML/AI/DL Developer

I am a Google Dialogflow expert, Chatbot developer, AI/ML/DL learner.

Ranjan Kumar Singh

Sr.Data Analyst at Amazon |
Mentor@Data Science

The individual is enthusiastic about learning new things and regularly updates themselves with the latest technologies.

Dr Jaya R

Data Science Faculty

18-year-old Associate Professor with expertise in education management, curriculum development, and faculty development. With a PhD in Computer Engineering, interests include Data Mining, Blockchain, and Data Science, and has authored books and papers.

Vasudev Gupta

Data Scientist at Decision Tree Analytics & Services

Incorporating a diverse range of expertise, our community specializes in statistical modeling, data mining, machine learning, and time series forecasting. We delve into cutting-edge techniques like deep learning and Bayesian modeling, while also emphasizing the practical applications through business intelligence and data visualization.

Satish Vavilapalli

Data Scientist at Fractal

With a PG Diploma In Business Analytics and over 6 years of experience in data science and analytics, the individual has expertise in Python, R, SQL, SAS, predictive analytic procedures, cross-validated ensemble models, predictive models, Time Series Analysis, and feature selection.

Dr. Suriya Begum

Sr. Corporate Trainer

The individual is a Ph.D. in Computer Science and Engineering, a data scientist with over 15 years of experience, and a dynamic trainer with excellent communication, management, and team-building skills. They have experience in designing and implementing various training programs, developing learning materials, and organizing workshops to help meet business goals.

Sai Ram Kasanagottu

B.Tech | M.S | IIT Kharagpur Alumnus

Experienced Research Fellow with expertise in real estate recommendation engine, unmanned aerial navigation, computer vision, deep learning, object recognition, remote sensing, medical image scan segmentation, and predictive diagnostics.

Rutvik Acharya

Data Science | Generative AI, LLMs & NLP | Atlassian | ex - Microsoft

Experienced Data Science, AI and Analytics professional with a demonstrated history of building data science products and data-driven value across industries like Technology, CPG and Automotive.

Lokabhiram Dwarakanath

Assistant Professor @ DSATM

A PhD candidate with expertise in Python Machine Learning and Deep Learning packages, with experience working on diverse datasets and ML algorithms like Naive Bayes, SVM, Linear Regression, Decision Tree, and Random Forest. Previously worked in Digital Marketing. Seeking a Data Scientist/Machine Learning Engineer role.

Rashmi Biswas

Machine Learning || Data Science

I possess strong analytical skills, specializing in analytics, machine learning, and NLP, and am eager to contribute to data management and contribute to a growth-oriented environment.

Raj Kapadia

ML/AI/DL Developer

A PhD candidate with expertise in Python Machine Learning and Deep Learning packages, with experience working on diverse datasets and ML algorithms like Naive Bayes, SVM, Linear Regression, Decision Tree, and Random Forest. Previously worked in Digital Marketing. Seeking a Data Scientist/Machine Learning Engineer role.

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UpShaala Placement Highlights

Job Portal

Our placement team curates job opportunities from leading companies that are suitable for your skillset, academic background and experience.

- 30+ Job opportunities every month
- Job openings tailored to your course relevance
- Industry guidance & interview tips from the alumni

Career Mentoring

We boost your potential to make a smart career decision. Get tailored and personalized career guidance from industry experts.

- 100% Live, Online & Interactive sessions with professional Industry Experts.
- Find the best career opportunities with excellent mentorship.
- Recommendations for establishing short-term and long-term career goals.
- Mentors from diverse fields assist you in gaining vital industry knowledge.

Interview Preparation

Get assistance in developing industry knowledge for getting interview ready. We polish your skills so that you are ready, groomed, and confident for your interview.

- Become interview ready with 1:1 mock interviews.
- Get Industry insights from our mentors, trainers and alumni panel.
- Prepare technical and HR questionnaires that are expected.

Portfolio Development

We help you create a portfolio that contains both industrial and personal work, as well as content that showcases your digital marketing skills and experience.

- Create an appealing portfolio with live assessment on projects, assignments and case studies.
- Develop your portfolio under the guidance and mentorship of our Industry Experts.

CV/LinkedIn Review

Build a solid CV and optimize your LinkedIn profile such that it stands out from the crowd.

- Expert guidance on making your CV and building your LinkedIn profile.
- Suggestions for tailoring your CV as per the job opening you're applying for.
- Recommendations for optimizing the structure and content of your CV/LinkedIn profile as per your course relevance.

Where Our Learners Work



Our Alumni's Thoughts



Ananya Gupta

NLP Engineer

Offered 21 LPA

Learning data science through this program was a game-changer for me. It expanded my career horizons and empowered me with skills that have been invaluable in my professional journey.



Vikram Singhania

Data Analyst

Offered 18 LPA

As a business analyst, mastering data science was essential. This course not only enhanced my analytical skills but also opened doors to exciting opportunities I never thought possible.



Priya Sharma

Generative AI Developer

Offered 14.5 LPA

I'm thrilled with the depth of knowledge gained in this data science course. It's given me a competitive edge and enabled me to contribute meaningfully to tech innovations in my company.



Rohan Patel

Chat GPT Expert

Offered 12 LPA

Studying data science here was a turning point. It helped me transition from a different field into a highly rewarding career. The course structure and mentorship were exceptional.



Nisha Verma

Deep Learning Engineer

Offered 11.5 LPA

This course demystified the complexities of data science, making it accessible and exciting. The hands-on experience was invaluable, shaping my journey as a data enthusiast.



Aarav Reddy

Data Engineer

Offered 10 LPA

Data science sounded intimidating, but this program broke it down into digestible chunks. Now, I'm confidently implementing data-driven strategies in my projects.

Internship Opportunity in Startups



Junior Data Scientist

Start working as Jr. Data science Intern in Analytics start-ups.



Hands-On Experience

Hands on Industry experience with 1 to 1 client engagement and intensive training with practical real time projects.



Certification

Gain Project Experience certificate from analytics Start-ups.



How this program operates?

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Students Journey



Real-Time Projects



Predictive Healthcare Analytics

Utilizing machine learning for early disease detection and personalized health insights, revolutionizing proactive healthcare.

[Read More](#)



Urban Infrastructure Optimization

Data-driven solutions for efficient urban planning and resource allocation, fostering sustainable development in smart cities.

[Read More](#)



Market Trend Forecasting Platform

Advanced algorithms providing real-time financial market predictions, empowering investors with insightful foresight.

[Read More](#)



Environmental Impact Analysis

Harnessing data science to evaluate environmental factors and their effects, driving informed decisions for eco-friendly initiatives.

[Read More](#)

Syllabus

Best-in-class content by leading faculty and industry leaders in the form of videos, cases and projects, assignments and live sessions

Module -1 AI-Enhanced Data Analysis with ChatGPT

Module-6 Learning Outcome :- (Job Role :- Chat GPT Expert)

After completing Module-6 you can start working as Chat GPT Expert :

Description:

This Module is designed to equip participants with the knowledge and skills to effectively use R programming, Python and ChatGPT for data analysis tasks. Data analysis, traditionally a human-driven process, can greatly benefit from the integration of AI and natural language processing technologies. In this course, participants will explore how to leverage ChatGPT as a valuable assistant in data interpretation, report generation, and automating repetitive data-related tasks.

Target Audience:

Aspirants who wanted to become Data analysts, data scientists, business analysts, and professionals involved in data analysis, who want to incorporate AI technologies, particularly ChatGPT, into their workflows to improve efficiency, generate insights, and enhance data-driven decision-making.

Introduction to R programming

- Overview of R and RStudio, Installation, R interface and basic commands
- Data Structures and Data Import
- Control Structures and Functions
- Data Visualization with ggplot2
- Descriptive statistics, Basic statistical tests

Introduction to Python Programming

- Introduction to Python
- Control Structures and Functions
- Data Structures in Python
- File Handling and Error Handling
- Object-Oriented Programming (OOP)

Introduction to Python Libraries for Data Analysis

- Introduction to NumPy
- Introduction to Pandas Series, Dataframes
- Index, Select, filter, Function Mapping, Sorting
- Dealing with Duplicate Values, Unique values, Missing values
- Data Visualization with Matplotlib

Data Analysis and ChatGPT

- Introduction to data analysis and ChatGPT
- Understanding data analysis tools and techniques
- ChatGPT architecture and applications
- Ethical considerations in data analysis and AI
- Setting up the development environment

Data Collection and Preprocessing

- Data collection methods and sources
- Data cleaning and preprocessing techniques
- Data transformation and feature engineering
- Dealing with missing data and outliers
- Introduction to ChatGPT for data preprocessing

Exploratory Data Analysis (EDA)

- Data visualization using data analysis tools
- Descriptive statistics and data summaries
- EDA best practices
- ChatGPT for generating EDA reports
- Practice and review

Statistical Analysis and Hypothesis Testing

- Understanding statistical concepts
- Common hypothesis tests and when to use them
- Interpreting statistical results
- ChatGPT for statistical analysis assistance
- Advanced hypothesis testing

Data Modeling and Machine Learning

- Introduction to machine learning
- Supervised and unsupervised learning
- Model evaluation and selection
- ChatGPT for model selection guidance
- Model deployment and monitoring

Advanced Data Analysis Techniques

- Time series analysis
- Natural language processing for text data
- Advanced feature engineering
- ChatGPT for generating advanced analysis reports
- Practice on real-world datasets

Integrating ChatGPT for Data Interpretation

- Customizing ChatGPT for data analysis tasks
- Data interpretation using ChatGPT
- Building custom AI-assisted dashboards
- ChatGPT for automated data insights
- Troubleshooting and error handling

ChatGPT in Industry-Specific Data Analysis

- Healthcare and medical data analysis
- Financial data analysis and forecasting
- E-commerce and market analysis
- Environmental data analysis
- Selected industry-specific case studies

Advanced Applications with ChatGPT

- ChatGPT in customer support
- ChatGPT in content generation and curation
- ChatGPT in e-learning and tutoring
- ChatGPT in research and data analysis
- Legal and Ethical Considerations

Module -2 Machine Learning Expert

Module-1 Learning Outcomes: – (Job Role :- Asso Data Scientist / Machine Learning Expert)
After completing Module-1 you can start working as Machine Learning Engineer

Introduction to Data Science.

Different Disciplines of Data Science:

- Machine Learning
- Natural Language Processing
- Deep Learning
- Computer Vision
- Generative AI
- ChatGPT
- Data Analytics
- Business Analytics
- Applications of Machine Learning

Why Machine Learning is the Future

- Creating the "Hello World" code
- Demonstrating Conditional Statements
- Variables
- Demonstrating Loops

Statistics & Probability

- Descriptive Statistics and Inferential Statistics
- Sample and Population
- Variables and Data types
- Percentiles
- Measures of Central Tendency
- Measures of Spread
- Skewness, Kurtosis
- Degrees of freedom
- Variance, Covariance, Correlation
- Descriptive statistics and Inferential Statistics in Python
- Test of Hypothesis
- Confidence Interval
- Sampling Distribution
- Standard Probability Distribution
- Functions
- Bernoulli, Binomial-Distributions
- Normal Distributions

Introduction to Python

- Overview of Python
- Different Applications where Python is used
- Values, Types, Variables
- Conditional Statements
- Command Line Arguments
- The Companies using Python
- Discuss Python Scripts on UNIX/Windows
- Operands and Expressions
- Loops
- Writing to the screen

Deep Dive – Functions, OOPs, Modules, Errors, and Exceptions

- Functions
- Global Variables
- Lambda Functions
- Standard Libraries
- The Import Statements
- Package Installation Ways
- Handling Multiple Exceptions
- Function Parameters
- Variable Scope and Returning Values
- Object-Oriented Concepts
- Modules Used in Python
- Module Search Path
- Errors and Exception Handling

Python in Data Manipulation

- Basic Functionalities of a data object
- Concatenation of Data objects
- Exploring a Dataset
- Merging of Data Objects
- Types of Joins on data Objects
- Analysing a Dataset
- Pandas Function- Ndim(), axes(), values(), head(), tail(),
- sum(), std(), iteritems(), iterrows(), itertuples()
- Aggregation
- Merging
- GroupBy operations
- Concatenation
- Joining

Sequences and File Operations

- Python files I/O Functions
- Strings and Related Operations
- Lists and Related Operations
- Sets and Related Operations
- Numbers
- Tuples and related operations
- Dictionaries and related operations
- File Operations using Python
- Working with data types of Python

Introduction to NumPy, Pandas and Matplotlib

- NumPy – Arrays
- Indexing Slicing and iterating
- Pandas – Data Structures & Index Operations
- Matplotlib Library
- Markers, Colours, Fonts and Styling
- Contour Plots
- Operations on Arrays
- Reading and Writing Arrays on Files
- Reading and Writing Data from Excel/CSV formats into
- Pandas
- Grids, Axes, Plots
- Types of Plots – Bar Graphs, Pie Charts, Histograms
- Probability Distributions in Python
- Python for Data Visualisation

Introduction to Machine Learning with Python

- What are Classification and its use cases?
- Algorithm for Decision Tree Induction
- Confusion Matrix
- What is Decision Tree?
- Creating a Perfect Decision Tree
- What is a Random Forest?
- Implementation of Logistic Regression
- Random Forest
- Decision Tree
- Introduction to Dimensionality
- PCA
- Scaling Dimensional Model
- Why Dimensionality Reduction
- Factor Analysis
- LDA
- What is Naïve Bayes?
- Implementing Naïve Bayes Classifier
- Illustrate how Support Vector Machine works?
- Grid Search vs Random Search
- How Naïve Bayes works?
- What is Support Vector Machine?
- Hyperparameter Optimization
- Implementation of Support Vector Machine for
- Classification
- Implementation of Naïve Bayes, SVM

Unsupervised Learning

- What is Clustering & its Use Cases?
- How does the K-means algorithm work?
- What is C-means Clustering?
- How does Hierarchical Clustering works?
- What is K-means Clustering?
- How to do Optimal clustering
- What is Hierarchical Clustering?
- Implementing K-means Clustering
- Implementing Hierarchical Clustering

Association Rules Mining and Recommendation Systems

- What are Association Rules?
- Calculating Association Rule Parameters
- How do Recommendation Engines work?
- Content-Based Filtering
- Association Rule Parameters
- Recommendation Engines
- Collaborative Filtering
- Apriori Algorithm

Reinforcement Learning

- What is Reinforcement Learning?
- Elements of Reinforcement Learning
- Epsilon Greedy Algorithm
- Q values and V values
- α values
- Why Reinforcement Learning
- Exploration vs Exploitation dilemma
- Markov Decision Process (MDP)

- Q - Learning
- Implementing Q - Learning
- Implement Reinforcement Learning using Python
- Developing Q - Learning model in Python

Time Series Analysis

- What is Time Series Analysis?
- Components of TSA
- AR Model
- ARMA Model
- Stationarity
- Importance of TSA
- White Noise
- MA Model
- ARIMA Model
- ACF & PACF
- Checking Stationarity
- Implementing the Dickey-Fuller Test
- Generating the ARIMA plot
- Converting a non-stationary data to stationary
- Plot ACF and PACF
- TSA Forecasting
- TSA in Python

Model Selection and Boosting

- What is the Model Selection?
- Cross-Validation
- How Boosting Algorithms work?
- Adaptive Boosting
- The need for Model Selection
- What is Boosting?
- Types of Boosting Algorithms

Cloud Computing & Deployment of Machine Learning Models to Cloud

- Flask basics
- Deployment of the model on Heroku
- AWS basics
 - S3
 - EC2
 - AWS Lambda
- Deployment of the model on EC2
- Deployment on AWS Lambda (Optional)
- Google Cloud Platform Basics
- Deployment of the Model on GCP
- Microsoft Azure basics
- Deployment of the Model on Azure
- Pyspark Basics
- DevOps Concepts

Module -3 Data Science Project Management Expert

Module-2 Learning Outcome :- (Job Role :- Data Science Project Manager)

After completing Module-2 you can start working as Data Science Project Lead / Manager

Data Science Project Management

- Leading Data Science Teams & Processes
- Exploring Methodologies
- How to manage data science projects and lead a data science team
- Agile Data Science
- Scrum Data Science
- Emerging Approaches – Microsoft TDSP
- Data Science Methodology understanding
- Business & Data understanding
- Modelling & Evaluation
- Plan Deployment
- Data Science Project Report

Module -4 NLP Expert

Module-3 Learning Outcome:- (Job Role :- NLP Expert)

After completing Module-3 you can start working as NLP Engineer

Introduction to Natural Language Processing

- Introduction to Natural Language Processing
- Understand and implement word2vec
- Understand the CBOW method in word2vec
- Understand the skip-gram method in word2vec
- Understand the negative sampling optimisation in word2vec
- Understand and implement GloVe using gradient descent and alternating least squares
- Use recurrent neural networks for parts-of-speech tagging
- Use recurrent neural networks for named entity recognition
- Understand and implement recursive neural networks for sentiment analysis
- Understand and implement recursive neural tensor networks for sentiment analysis

Natural Language Processing in TensorFlow

- Build natural language processing systems using TensorFlow
- Process text, including tokenisation and representing sentences as vectors
- Apply RNNs, GRUs, and LSTMs in TensorFlow
- Train LSTMs on existing text to create original poetry and more

First Step of NLP – Text Processin

- Build natural language processing systems using TensorFlow
- Process text, including tokenisation and representing sentences as vectors
- Apply RNNs, GRUs, and LSTMs in TensorFlow
- Train LSTMs on existing text to create original poetry and more

Extracting Named Entities from Text

- Understanding Named Entity Recognition
- Exercise: Understanding Named Entity Recognition
- Implementing Named Entity Recognition
- Exercise: Implementing Named Entity Recognition
- Named Entity Recognition and POS tagging using spaCy
- POS and NER in Action: Text Data Augmentation
- Assignment: Share your learning and build your profile

Feature Engineering for Text

- Introduction to Text Feature Engineering
- Count Vector, TFIDF Representations of Text
- Exercise: Introduction to Text Feature Engineering
- Understanding Vector Representation of Text
- Exercise: Understanding Vector Representation of Text
- Understanding Word Embeddings
- Word Embeddings in Action - Word2Vec
- Word Embeddings in Action - GloVe

Mastering the Art of Text Cleaning

- Introduction to Text Cleaning Techniques Part 1
- Exercise: Introduction to Text Cleaning Techniques Part 1
- Introduction to Text Cleaning Techniques Part 2
- Exercise: Introduction to Text Cleaning Techniques Part 2
- Text Cleaning Implementation
- Exercise: Text Cleaning Implementation
- NLP Techniques using spaCy

Interpreting Patterns from Text - Topic Modelling

- Introduction to Topic Modelling
- Exercise: Introduction to Topic Modelling
- Understanding LDA
- Exercise: Understanding LDA
- Implementation of Topic Modelling
- Exercise: Implementation of Topic Modelling
- LSA for Topic Modelling

Understanding Text Classification

- Overview of Text Classification
- Exercise: Overview of Text Classification
- Assignment: Share your learning and build your profile

Introduction to Language Modelling in NLP

- Overview: Language Modelling
- What is a Language Model in NLP?
- N-gram Language Model
- Implementing an N-gram Language Model - I
- Implementing an N-gram Language Model - II
- Neural Language Model
- Implementing a Neural Language Model

Sequence-to-Sequence Modelling

- Intuition Behind Sequence-to-Sequence Modelling
- Need for Sequence-to-Sequence Modelling
- Understanding the Architecture of Sequence-to-Sequence
- Understanding the Functioning of Encoder and Decoder
- Case Study: Building a Spanish to English Machine Translation Model
- Pre-processing of Text Data
- Converting Text to Integer Sequences
- Model Building and Inference

Advanced NLP Tools

- Text Classification & Word Representations using FastText (An NLP library by Facebook)
- Introduction to Flair for NLP: A Simple yet Powerful State-of-the-Art NLP Library
- Introduction to Stanford NLP: An Incredible State-of-the-Art NLP Library for 53 Languages (with Python code)
- A Step-by-Step NLP Guide to Learn Elmo for Extracting Features from Text
- Tutorial on Text Classification (NLP) using ULMFiT and fastai Library in Python
- 8 Excellent Pretrained Models to get you started with Natural Language Processing (NLP)
- Geo-coding using NLP by Shantanu Bhattacharyya and Farhat Habib
- Demystifying the What, the Why and How of Chatbot by Sonny Laskar
- Sentiment Analysis using NLP and Deep Learning by Jeeban Swain
- Identifying Location using Clustering and Language Model - By Divya Choudhary
- Building Intelligent Chatbots from Scratch

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Module -5 Deep Learning Expert

Module-4 Learning Outcome :- (Job Role :- DEEP LEARNING Expert)

After completing Module-4 you can start working as DEEP LEARNING Engineer

Introduction to Deep Learning

Getting started with Neural Network
Exercise: Getting started with Neural Network
Understanding Forward Propagation
Exercise: Forward Propagation
Math Behind forwarding Propagation
Exercise: Math Behind forwarding Propagation
Error and Reason for Error
Exercise: Error and Reason for Error
Gradient Descent Intuition
Understanding Math Behind Gradient Descent

- Exercise: Gradient Descent
- Optimiser
- Exercise: Optimiser
- Back Propagation
- Exercise: Back Propagation
- Why Keras?
- Exercise: Why Keras?
- Building a Neural Network for Text Classification
- Why CNN?
- Exercise: Why CNN?
- Understanding the working of CNN Filters
- Exercise: Understanding the working of CNN Filters
- Introduction to Padding
- Exercise: Introduction to Padding
- Padding Strategies
- Exercise: Padding Strategies
- Padding Strategies in Keras
- Exercise: Padding Strategies in Keras
- Introduction to Pooling
- Exercise: Introduction to Pooling
- CNN architecture and it's working
- Exercise: CNN architecture and it's working

Deep Learning for NLP

- Deep Learning for NLP Part 1
- Exercise: Deep Learning for NLP Part 1
- Deep Learning for NLP Part 2
- Exercise: Deep Learning for NLP Part 2
- Text Generation Using LSTM
- Exercise : Text Generation Using LSTM

Recurrent Neural Networks

- Why RNN
- Introduction to RNN: Shortcomings of an MLP
- Introduction to RNN: RNN Architecture
- Training an RNN: Forward propagation
- Training an RNN: Backpropagation through time
- Need for LSTM/GRU
- Long Short Term Memory (LSTM)
- Gated Recurrent Unit (GRU)
- Project: Categorisation of websites using LSTM and GRU I
- Dataset and Notebook
- Project: Categorisation of websites using LSTM and GRU II

Module -6 Generative AI Developer (Basics)

Module-5 Learning Outcome :- (Job Role :- Generative AI Expert)

After completing Module-5 you can start working as GENERATIVE AI EXPERT

- Getting Started with Google Bard
- Getting Started – AI vs ML vs Generative AI
- Exploring Generative AI Fundamentals
- NIL in Google Cloud -An Overview
- Generative AI with Google and Google Cloud – Quick Overview Getting Started with Prompt Design
- Exploring Vertex AI PaLNI API and Generative AI Studio Further Exploring Generative AI Further
- Exploring Other Google Generative AI Offerings
- Google AI Principles and Responsible AI practices
- Generative AI with Google–Congratulations

Module- 7 GENERATIVE AI Developer (Advance)

Sequence-to-Sequence Modelling

Class 1-2: Overview of Generative AI.

- Introduction to Generative AI and its applications
- Understanding the basics of machine learning and deep learning

Class 3-4: Neural Networks Fundamentals

- Overview of neural networks
- Activation functions, loss functions, and optimization algorithms

Class 5-6: Introduction to Natural Language Processing (NLP)

- Basics of NLP
- Tokenization, stemming, and lemmatization

Class 7-8: Introduction to TensorFlow and PyTorch

- Setting up the development environment
- Basic operations and building simple models

Month 2: Deep Dive into Generative Models

Class 9-10: Introduction to Generative Models

- Understanding generative models: GANs, VAEs
 - Use cases and applications
- #### Class 11-12: GANs (Generative Adversarial Networks)
- Working principles of GANs
 - Building and training GANs using TensorFlow and PyTorch
- #### Class 13-14: VAEs (Variational Autoencoders)
- Understanding VAEs
 - Implementing VAEs for generative tasks
- #### Class 15-16: Advanced Topics in Generative Models
- Conditional GANs, WGANs, and other advanced architectures
 - Transfer learning in generative models

Month 3: Prompt Engineering and Fine-tuning

Class 17-18: Introduction to Prompt Engineering

- Overview of prompt engineering in Generative AI
- Designing effective prompts for desired outputs
- Prompt Engineering- Zero Shot vs Few Shot Learning

Class 19-20: Fine-tuning Models

- Transfer learning and fine-tuning pre-trained models
- Hands-on practice with fine-tuning
- Limitations of LLMs: How to overcome with limitations of LLMs with Retrieval Augmented Generation(RAG) with LangChain: Fine-tuning

Class 21-22: Evaluation Metrics for Generative Models

- Understanding metrics for evaluating generative models
- Precision, recall, F1 score, and other relevant metrics
- Corresponding Metrics - Micro F1; Macro F1; ROUGE-L; Exact Match; BLUE ROUGE-L

Month 4: Real-world Applications and Project Work

Class 23-25: Applications of Generative AI in Industry

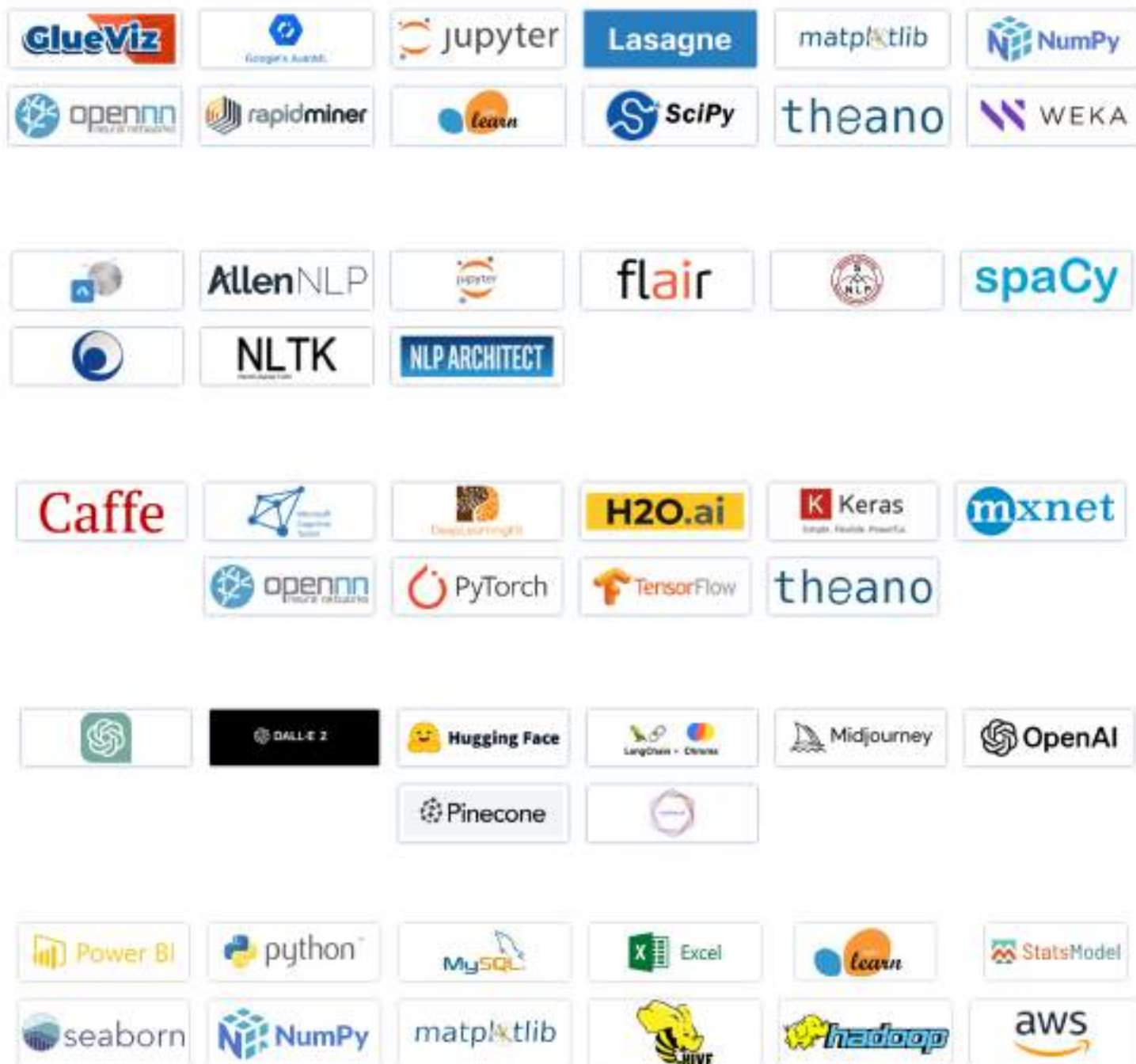
- Real-world use cases in various industries
 - Ethical considerations in generative AI
 - How can managers and corporates can increase productivity with Generative AI
 - Applications of Generative AI in various industries with special focus on Generative AI in Financial Services
 - Generative AI applications in Finance- Case studies (Q&A on Financial Disclosures and Fundamental Analysis using LLMs on 10Ks)
 - Creating domain specific Chatbots Using LangChain
 - Content Creation- Text Generation (News article generation given a headline; Summarisation of RBI speeches)
 - Sentiment Analysis with few shot prompting
- #### Class 26-28: Project Kickoff and Guidance
- Students select and start working on their individual projects
 - Guidance on project scope, implementation, and documentation

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